

Unit - IV

Q) Classify, advantage and disadvantage and preformulation factors of Parenteral Products?

A) Parenteral (para = beyond; enteral = intestine) dosage forms differ from all other drug dosage forms because they are injected directly into body tissues through the primary protective systems of the Human Body: Skin, Mucous Membrane.

→ They must be exceptionally pure and free from physical, chemical and biological contaminants

Classification of Parenteral Products

① Based on Volume

- Small Volume Parenteral (Max^m volume upto 100ml)
- Large Volume Parenteral (Max^m volume more than 100ml)

② Based on type of Packaging

- Single Dose - No Need of Preservative (eg Infusion)
- Multiple Dose - Contain Preservative (eg Eye Drops) (Upto 30ml)

③ Based on Physical State //

- Solution (Most Commonly Used)
- Powder (Should be reconstituted before injecting with water for injection)
- Suspension (Particle should be sufficiently small)

Advantage :-

- They also have essential 100% Bioavailability & can be used for drugs that are poorly absorbed or ineffective
- Some medication, such as certain antipsychotics can be administered via long-acting intramuscular injection
- Accurate Dosing and Minimized First Pass Effect
- Can be administered to unconscious patients, infants, elderly person and patient who cannot take oral medication

Disadvantages

- Injection include potential pain or discomfort for the patient & the requirement of Trained staff using aseptic techniques for administration
- As the drug is delivered to the site of injection rapidly with i/v injection, there is a risk of overdose if the dose has been calculated incorrectly.

Preformulation Factors Affecting Parenteral Products

- Preformulation is the study of physical and chemical properties of drug prior to formulation.
- These studies are performed under stressed condition of temperature, Humidity, light and oxygen so that the reaction are accelerated and potential reaction can be detected
- Various Physicochemical properties which affect a drug substance are:-

① pH

- Drug formulation should have a target pH as close as possible to physiological pH.
- The acceptable range is pH 2-11 for i/v and pH 4-9 for subcutaneous injection due to potential irritation issues

② Isotonicity

- Calculations are most often performed for parenteral and ophthalmic solution which must have freezing point depression of 0.52°C for them to be isotonic with Blood Plasma and Tears

③ Odour

- The odour of a new drug substance is examined by cautiously smelling the Headspace of the drug container.
- The presence and description of any odor is recorded

④ Molecular Structure and Weight

- Molecular structure and weight are the basic characteristics of the drug which must be determined

⑤ Particle Size and Shape

These characteristics are usually determined by microscopic evaluation by using a scanning electron microscope or an optical microscope with polarizing attachments

⑥ Melting Point

Melting Point determination is a primary indication of purity since the presence of relatively small amount of impurity can be detected by a lowering and widening in the MP range

⑦ Hygroscopicity

Hygroscopicity is the phenomenon of absorption of moisture by compounds under specific conditions of moisture and Humidity

⑧ Partition Coefficient

- The partition coefficient P is a measure of lipophilicity of a compound

$$P = [C_{oil}] / [C_{water}]$$

Q3) Write down the Formulation Approaches of Parenterals?

A3) Excipients used in the Parenteral Preparation must be non-pyrogenic, non-toxic, non-Hemolytic and non-irritating should be physically and chemically compatible with API must not interfere with the therapeutic effect of the Active Ingredients must ~~not~~ maintain stability during sterilization and during the shelf life of the Product

(i) Vehicles

→ Vehicles are the liquid phase used in formulation of parenteral

① Water for Injection (WFI)

- Water for injection is Highly purified water which is subsequently sterilized and used as vehicle for the purpose of Injectable Preparation
- PH of the WFI is 5.0 to 7.0

② Sterile Water for Injection

- This is also known as sterile water for irrigation
- It is used for washing wounds, surgical incision or body tissue

(ii) Anti-Microbial Agents

The USP states that ^{anti}microbial agents in Bacteriostatic or Fungistatic concentrations must be added to preparations contained in Multiple-Dose containers.

(iii) Buffers

Buffers are used primarily to stabilize a solution against chemical degradation or especially for physical degradation (i.e. aggregation and precipitation)
eg Acetic acid & Salt

(iv) Tonicity Agents

Tonicity Agents are used in many parenteral and ophthalmic product to adjust the Tonicity of the solution
eg NaCl - 0.9% ; Dextrose - 5%.

(v) Suspending Agents

They are excipients which are added to the formulation in order to improve the stability of the formulation by preventing the sedimentation of the particles.

(vi) Antioxidants

They are added to the formulation to prevent the process of oxidation therefore increasing the stability of the formulation
eg BHT, BHA

(vii) Co-solvents

Co-solvent agents increase the solubility of drug by ~~water~~ disrupting the Hydrophobic interaction at the non-polar solute/water interface
eg PEG, Sorbitol

Q) Write down the Production Procedure of Parenteral Preparation?

A) Steps Involved In Production are :-

① Cleaning of Container, Closure and Equipments

→ All the containers, closures and equipments requires in preparation of parenteral thoroughly cleaned with detergent and distilled water

→ Rubber closures are washed with Hot solution of .5% Sodium Pyrophosphate in water, removed from solution, washed with water and rinsed with Water for Injection

② Collection of Material

* Required Quantity of Ingredients for formulation weighed & collected in Preparation Room

* Water for Injection should be free from pyrogen microorganism

③ Preparation of Parenteral Product

→ The pharmacist should decide order of mixing and exact method of preparation

→ The parenteral preparation must be prepared under strict aseptic condition

④ Filtration

→ The parenteral solution so formed is pass through a bacteria proof filter, filter candle, Sertz filter, Membrane filter objective is to clarify solution

5) Filling the Preparation In final Container

- * Filtered Product is filled into Final Container
- * Ampoules - Single Dose Vials - Multi Dose
- Transfusion Bottles - Transfusion Fluid

6) Sealing the Container

- Ampoules are sealed manually by rotating neck of ampoule in flame of candle
- Vials and Transfusion bottles are sealed by closing with rubber closures

7) Sterilization

Parenteral Preparation should be immediately sterilized after sealing in Final Container

Thermostable Medicament
(Auto claving & Hot Air oven)

Thermolabile medicament
(Should be sterilized before sealing)

8) Labelling & Packaging

- After evaluation → ampoules, vials & transfusion bottles are properly labelled & packaged

Q2) How are Dry Powder for Injection Prepared by Lyophilization?

A2) Lyophilization is the process of drying in which water is removed by freezing and when put under vacuum the ice directly converted into vapour without passing through liquid phase (Sublimation)

Principle :- Lyophilization is based on sublimation (Transition of Solid Directly Into Vapours)

Objective of Lyophilization

- ① To dry Thermolabile materials
- ② To extend stability/shelf life of Product
- ③ To get accurate, sterile amount of product

Steps Involved In Lyophilization includes following :-

- Dissolving the drug & excipients in a suitable solvent; generally Water for Injection
- Sterilizing the Bulk Solution by passing it through a 0.22 micron Bacteria Retentive Filter
- Filling into individual sterile containers and partially stoppering the container under aseptic condition

- Transporting the Partially stoppered Container to the lyophilizer and loading into the chamber under aseptic condition
- Applying a vacuum to the chamber and Heating the ^{ly}phers in order to evaporate the water from the frozen state
- Complete stoppering of the vials usually by Hydraulic or screw rod stoppering mechanism installed in the lyophilizer

Advantage of Lyophilization

- ① Water is removed without Heat which prevents product degradation
- ② Shelf life of the drug product is extended and stability is enhanced
- ③ Product is processed in an aseptic condition

Disadvantage of Lyophilization

- ① Equipment and processes required for lyophilization are expensive to acquire and maintain
- ② The process can be time consuming taking over 24 Hours on average
- ③ It require Sterile Diluents for Reconstitution

① Quality Control Test of Parenteral?

A) Leakage Test

Method I

- Performed by immersing the ampoules in a vacuum chamber containing dye (1% Methylene Blue solution)
- A vacuum (about 27 inch Hg) created for about 15-30 min
- Vacuum causes the dye solution enter into ampoules with defective sealing
- If solution in Ampoules appear blue it confirms leakages

Method II

High Frequency Spark test to detect pin Holes in the Ampoules

- The machine uses High Precision Electrodes
- All containers presented individual to electrodes
- All product with Higher Voltage than defined max^m value are separated

② Clarity Test

This test is performed to detect presence of particulate matter in finished parenteral product

LVP contains

- NMT 50 particles of 10 micrometer/ml
- NMT 5 particles of 25 micrometer/ul

SVP contain

- NMT 10,000 particles of 10 micrometer/ml
- NMT 1000 particles of 25 micrometer/ml

(a) Visual Method

- Filled containers are examined by holding their neck against black and white background
- The container are slowly rotated and inverted all contents are examined for foreign particles

(b) Microscopic Count Method

A measured sample is filtered through membrane filter

↓
surface of filter examined under magnification of about 100X under microscope

3) Pyrogen Test

- Pyrogen are fever producing metabolic product of microorganisms

@ Rabbit Test

- Test carried out on a group of 3 rabbits

Procedure

Dissolve test substance or dilute it with pyrogen free saline



Insert a clinical thermometer into rectum of each rabbit & record temperature



Test solution is injected into ear vein of each rabbit



Record the temp of each rabbit at an interval of 30 min for 3 hours

Result

- If response of all rabbit is less than 0.6°C and if sum response of 3 rabbits is not more than 1.4°C



Then preparation pass the Test

- If not pass



then same test is repeated on another set of 5 rabbits



if not more than 3 out of 8 rabbits shows individual responses of 0.6 or more & if sum of responses of rabbits is not more than 3.7°C



Then preparation pass the test

③ Limulus Amoebocyte Lysate (LAL) Test

→ The test based on formation of gel or development in presence of Bacterial Endotoxin and lysate solⁿ

Procedure

The lysate solⁿ is mixed with equal Vol^m of test solⁿ in a pyrogen free test tube



Allowed to stand for about 60 min



Tube Inverted for observation of Formation of Gel Clot



If solid gel not form indicate absence of ~~endotoxin~~ ^{endotoxin}



Solution Pass the test

④ Sterility Test

This test is used to identify presence or absence of viable microorganism in preparation

(i) Direct Incubation Method

Quantity of Test substance in each culture medium is directly transferred under aseptic condⁿ



The medium is incubated

Soyabean Casein digest medium - at 20-25°C } for 14 days
Fluid Thioglycollate Medium - at 30-35°C }

⑥ Membrane Filter Method

→ Sterile membrane filter of pore size not greater than $0.45 \mu\text{m}$ is used

→ The test solution filtered through membrane ^{filter} under aseptic condition

↓
Then membrane cut into two equal portions

↓
One Half is immersed in 100ml soyabean - Casein digest medium

(incubated at $20-25^\circ\text{C}$) → not less than 7 days

↓
Other Half immersed in 100ml fluid thioglycollate medium
(incubated at $30-35^\circ\text{C}$) → not less than 7 days

Q) ^hOphthalmic Preparation; its Advantage and Disadvantage

A) Ophthalmic Preparation are sterile solutions, gel or ointments that are used to prevent, treat or protect the eyes from infection or foreign particles

Advantages

- More stable formulation
- Not causing Blurry Vision after application
- Rapid onset of action
- Convenient for patients who have difficulty applying ointment

Disadvantage

- The very short time the solution stays at the eye surface
- To be administered at frequent intervals
- Not individual fit for each patient

Q) Formulations and Methods of Preparation of Eye Drops?

A) (i) Antioxidants

- By preventing their oxidation during storage they reduce the degradation of ophthalmic solution or eye drops
- The preparation can be protected from oxidation by replacing the air with inert gas like Nitrogen
eg Sodium Thiosulphate (0.1 to 0.2%)

(ii) Buffers

- The pH of the preparation should be selected such that
 - should not destabilize or decrease solubility of drug
 - should not cause any ocular irritation
- They are used to maintain the pH of solⁿ at desired level.
eg Borate Buffer, Phosphate Buffer

3) Chelating Agents

- Chelating Agents are used in ophthalmic preparation as they form complexes with metals and prevent the oxidation of drugs

eg Disodium Edetate (0.01 - 0.3%)

4) Preservative

- Preservative are added in eye drops to prevent growth of microorganism which may have accidentally enter during repeated usage.

- So multi dose containers must contain a suitable preservative

eg Phenylmercuric Acetate and Nitrate

5) Thickening Agents (Viscosifiers)

- These are used to increase the contact time b/w drug and eye so increase therapeutic action

eg Hydroxy propyl Methyl Cellulose (HPMC)
Polyethylene Glycol

6) Isotonicity Adjustment Substance

- Eye drops are made isotonic with the lacrimal secretions

eg NaCl solution

7) Vehicles

Special vehicles are available for use in ophthalmic

eg Boric Acid Vehicles

8) Wetting Agents

These are used for proper penetration of eye drops into cornea of the eye

eg Polysorbates 20 and Polysorbate 80

Method of Preparation

1) Preparation of Vehicle

→ In the preparation of eye drops we first prepare a first preservative vehicle
(stock solⁿ are avoided as they susceptible to undergo chemical changes)

2) Dissolution of the Ingredients

→ Weigh the required quantity of ingredients carefully by preventing any atmospheric contamination

Add drug to the vehicle and dissolve it completely
↓ dissolve remaining ingredients

Adjust the pH of the preparation with a solⁿ of definite strength
↓

Make up the final volume & adjust & check the pH

3) Clarification

The solution is filtered through
↓

Bacteria Proof Filters like Seitz Filters, Membrane Filter (0.45 to 1.2 μ m pore size) under aseptic condition
↓

→ The clarified filtrate is filled directly into the containers which are then sealed.

4) Sterilization

1) Autoclaving

• Solution in final container Heated in an Autoclave
115°C for 30min at 15 lbs Pressure

2) Heating

The solution is Heated in an Apparatus called Steamer
at 98-100°C for 30min

3) Filtration

Used for Thermolabile drugs solution is filtered
through sterile membrane filter with pore size
• $0.45 \pm 0.02 \mu\text{m}$ under aseptic condition

Q) Quality Control of Ophthalmics Preparation

A) Leakage Test ; Sterility Test ; Clarity Test

4) Metal Particles in Ophthalmic Preparation

Extrude completely the content of 10 containers
separately into 10 flat bottom Petri dishes

↓
Cover Heat at 85°C for 2 hrs cool & slow to modify

↓
Remove the covers and insert on the stage of microscope
equipped with eye piece micrometer disk calibrated
for magnification

↓
Examine the entire bottom of Petri Dish for
Metal Particles

↓
Count the particles of 50µm or larger

Result

Test is pass if → number of such particle in all 10 tube does not exceed 50 & not more than one tube contain more than 8 such particles

↓ if not

Repeat with 20 more tubes

↓
Test is pass if total number of such particles in 30 tube is not more than 150 and not more than 3 tubes contains more than 8 particles